Dev Kumar Dahiya

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PROFESSIONAL SUMMARY

Computer Science graduate (CGPA 8.5) with hands-on experience designing and deploying Generative AI and data-driven solutions using Python, LLMs, and cloud platforms. Built production-ready AI pipelines for document intelligence and analytics, achieving high accuracy and measurable business impact. Experienced in data pipelines, REST APIs, and cloud-based AI systems (Azure, Databricks, Snowflake). Strong ability to translate business problems into scalable AI solutions.

TECHNICAL SKILLS

AI / Generative AI:
LLMs, Prompt Engineering, GROQ, Hugging Face (familiar), NLP, Transformers, Embeddings, Model Evaluation

Machine Learning & Deep Learning:
PyTorch, Scikit-learn, CNN, BiLSTM, Attention Mechanisms, Transfer Learning

Data Engineering & Analytics:
ETL Pipelines, Data Processing, Pandas, NumPy, Feature Engineering, Power BI, Tableau

Cloud & Platforms:
Microsoft Azure, Azure Data Factory, Databricks, Snowflake

Backend & APIs:
Python, REST APIs, C#, ASP.NET Core, Node.js

Tools:
Git, GitHub, Streamlit, Jupyter, Google Colab

WORK EXPERIENCE

Horizon 17 Technologies | *May 2025 – Jul 2025* | *Remote*

- Designed and delivered an end-to-end Generative AI pipeline to analyze invoices and estimate carbon emissions, integrating GROQ-powered LLMs for structured and unstructured document understanding — achieving ~90% field-level extraction accuracy and translating a fully manual process into an automated AI-driven solution.
- Applied NLP techniques — including tokenization, named entity recognition (NER), and prompt engineering — to extract key financial and environmental data fields from varied PDF formats, demonstrating practical ability to build and tune NLP-based text analytics applications.
- Evaluated and validated model outputs against ground-truth data, iterating on prompt design and preprocessing steps to improve extraction reliability across diverse document structures.
- Translated model results into standardized CO2e business metrics, saving analysts ~3 hours per report and enabling data-driven sustainability reporting — collaborating with cross-functional stakeholders to align AI outputs with business requirements.
- Built scalable data processing workflows aligning with enterprise AI solution architecture principles

Technologies: *Python, GROQ, LLMs, Prompt Engineering, NLP, PyTorch, Pandas, NumPy, Matplotlib, Streamlit,GROQQ API*

Software Engineer Intern (Associate) | MAQ Software *Nov 2025 – Feb 2026 | Noida*

Developed and optimized **data-driven solutions and dashboards**, reducing manual reporting workload by ~40% and enabling real-time decision-making for business stakeholders.

Built and automated ETL pipelines using **Azure Data Factory and Databricks**, enabling faster and scalable data processing.

Technologies: Python, C#, ASP.NET Core, REST APIs, Power BI, Microsoft Azure, Azure Data Factory, SQL Server, Git,Databricks,Snowflake,Rest.API

AI / GEN AI PROJECTS

Emotion-Based Music Recommendation System link *Oct 2025 – Dec 2025*

- Built and evaluated a deep learning model (CNN) for real-time facial emotion recognition on the FER-2013 dataset, achieving 85% test accuracy across 7 classes — applying model evaluation metrics and optimization techniques (data augmentation, dropout regularization, learning rate scheduling) to maximize generalization.
- Implemented a data preprocessing and feature engineering pipeline for image data, improving model input quality and contributing to a 12-percentage-point accuracy gain over the baseline.
- Deployed the AI solution as an interactive Streamlit application with real-time inference at 28 FPS and sub-200ms end-to-end latency, demonstrating ability to support deployment and monitoring of AI models.

Technologies: Python, PyTorch, CNN, OpenCV, NumPy, Streamlit, Scikit-learn, Machine Learning

Leave Request Management System link *Jan 2026 – Feb 2026*

- Developed and deployed a full-stack application with RESTful APIs and role-based access control, demonstrating ability to build, document, and ship end-to-end software solutions in a fast-paced development cycle.
- Integrated cloud data storage (MongoDB Atlas) and deployed across Vercel and Render, achieving sub-2-second load times — showing familiarity with cloud platforms and API-based architectures relevant to AI model serving.

Technologies: React.js, Node.js, Express.js, MongoDB, JWT, REST API, Git

RESEARCH & PUBLICATIONS

Forensic Deepfake Audio Detection Using CNN-Attention Mechanism | ICICC 2026 | Feb 2026

- Researched and developed a Generative AI detection model combining CNN feature extraction, BiLSTM sequence modeling, and multi-head attention mechanism — achieving 99.33% classification accuracy across 5+ accent groups, published at the International Conference on Intelligent Computing and Communication (ICICC 2026).
- Engineered an audio feature preprocessing pipeline (mel-spectrogram, MFCC, embeddings) and conducted rigorous model evaluation across diverse datasets, reflecting core competencies in AI research, model optimization, and result documentation.

EDUCATION

Bachelor of Technology — Computer Science & Engineering | Amity University | 2022 – 2026 | Noida, UP

CGPA: 8.5 / 10.0

CORE COMPETENCIES

- AI Solution Development
- Data Pipeline Design
- Cloud-Based AI Systems
- Business Problem Solving with AI
- API Integration & System Design